# DoubleMLDeep: Estimation of Causal Effects with LLMs and Multimodal Data

Ausschuss für Ökonometrie

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# Motivation

# Motivation

| | A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 | X9 | X10 | Y |
| 2 | 0,26222194 | 0,73438599 | 0,44667203 | 0,97138377 | 0,081064 | 0,28441456 | 0,30204598 | 0,42916989 | 0,45202869 | 0,48458485 | 0,69688241 |
| 3 | 0,02086748 | 0,76377164 | 0,82984213 | 0,44114913 | 0,6791843 | 0,10440879 | 0,06871173 | 0,65049866 | 0,38145381 | 0,37883627 | 0,68724919 |
| 4 | 0,80214042 | 0,60108851 | 0,3924631 | 0,64680441 | 0,52447716 | 0,71441139 | 0,92058036 | 0,50487157 | 0,0639613 | 0,72018174 | 0,70586344 |
| 5 | 0,69860016 | 0,48544746 | 0,02697306 | 0,56612716 | 0,64030572 | 0,38122836 | 0,77357736 | 0,7665354 | 0,56153968 | 0,40036439 | 0,96259876 |
| 6 | 0,76871281 | 0,52693951 | 0,23152259 | 0,99343286 | 0,19512117 | 0,52281687 | 0,74631891 | 0,51990814 | 0,8916446 | 0,13854693 | 0,15014547 |
| 7 | 0,80559982 | 0,49248903 | 0,52094313 | 0,83491804 | 0,19224862 | 0,51690107 | 0,99896405 | 0,21315395 | 0,13844972 | 0,5972433 | 0,30108879 |
| 8 | 0,14274622 | 0,68082916 | 0,21375762 | 0,12917741 | 0,101114 | 0,35962408 | 0,37101036 | 0,88481715 | 0,0917219 | 0,81144147 | 0,23748577 |
| 9 | 0,5565959 | 0,27805181 | 0,38191987 | 0,96746964 | 0,88262331 | 0,64204409 | 0,61391038 | 0,14798767 | 0,75991882 | 0,71364563 | 0,26831957 |
| 10 | 0,86680082 | 0,98940603 | 0,08032276 | 0,51284266 | 0,85067788 | 0,31879029 | 0,14212956 | 0,58889047 | 0,85003262 | 0,40984155 | 0,90320049 |
| 11 | 0,49294831 | 0,9930191 | 0,32376346 | 0,76486908 | 0,71150197 | 0,49307446 | 0,00732655 | 0,1366301 | 0,95413681 | 0,8192993 | 0,41659136 |
| 12 | 0,83367503 | 0,99203204 | 0,94500561 | 0,38442651 | 0,08267135 | 0,44319059 | 0,50597368 | 0,33789173 | 0,71150674 | 0,71723848 | 0,82498573 |
| 13 | 0,30901589 | 0,45442227 | 0,39182182 | 0,94058761 | 0,84662039 | 0,13237356 | 0,39203009 | 0,23863556 | 0,66904622 | 0,66739594 | 0,05703788 |
| 14 | 0,84854041 | 0,07220964 | 0,60564451 | 0,11235493 | 0,11423158 | 0,19425556 | 0,10525554 | 0,69400453 | 0,24273427 | 0,74521656 | 0,1934563 |
| 15 | 0,41204349 | 0,37890159 | 0,21172493 | 0,47214042 | 0,28972671 | 0,28045945 | 0,26144942 | 0,50182126 | 0,56848988 | 0,34171099 | 0,55046509 |
| 16 | 0,8270131 | 0,7550563 | 0,68080658 | 0,99718432 | 0,54748336 | 0,07330951 | 0,96389321 | 0,43492645 | 0,32444209 | 0,37929049 | 0,54724228 |
| 17 | 0,65968289 | 0,42809072 | 0,50651859 | 0,85136778 | 0,67725248 | 0,80776288 | 0,80154682 | 0,41961695 | 0,03794024 | 0,10550978 | 0,18250098 |
| 18 | 0,00250478 | 0,10976101 | 0,58665193 | 0,59539311 | 0,06932994 | 0,44532097 | 0,38697059 | 0,86107306 | 0,32445673 | 0,03897245 | 0,4593923 |
| 19 | 0,1646558 | 0,11563336 | 0,54131562 | 0,70212986 | 0,78935386 | 0,65759136 | 0,96083497 | 0,18585388 | 0,02245136 | 0,08388247 | 0,9684134 |

# Motivation

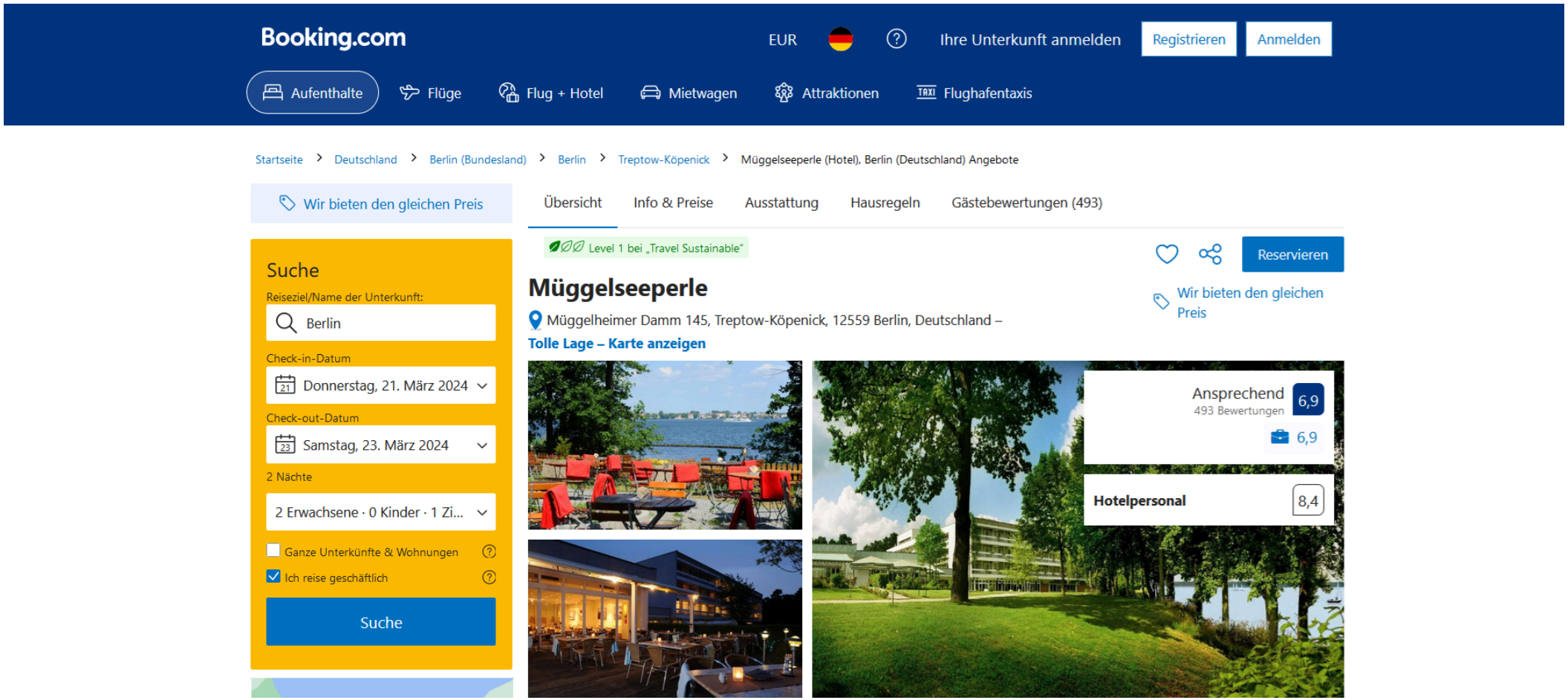


# Motivation

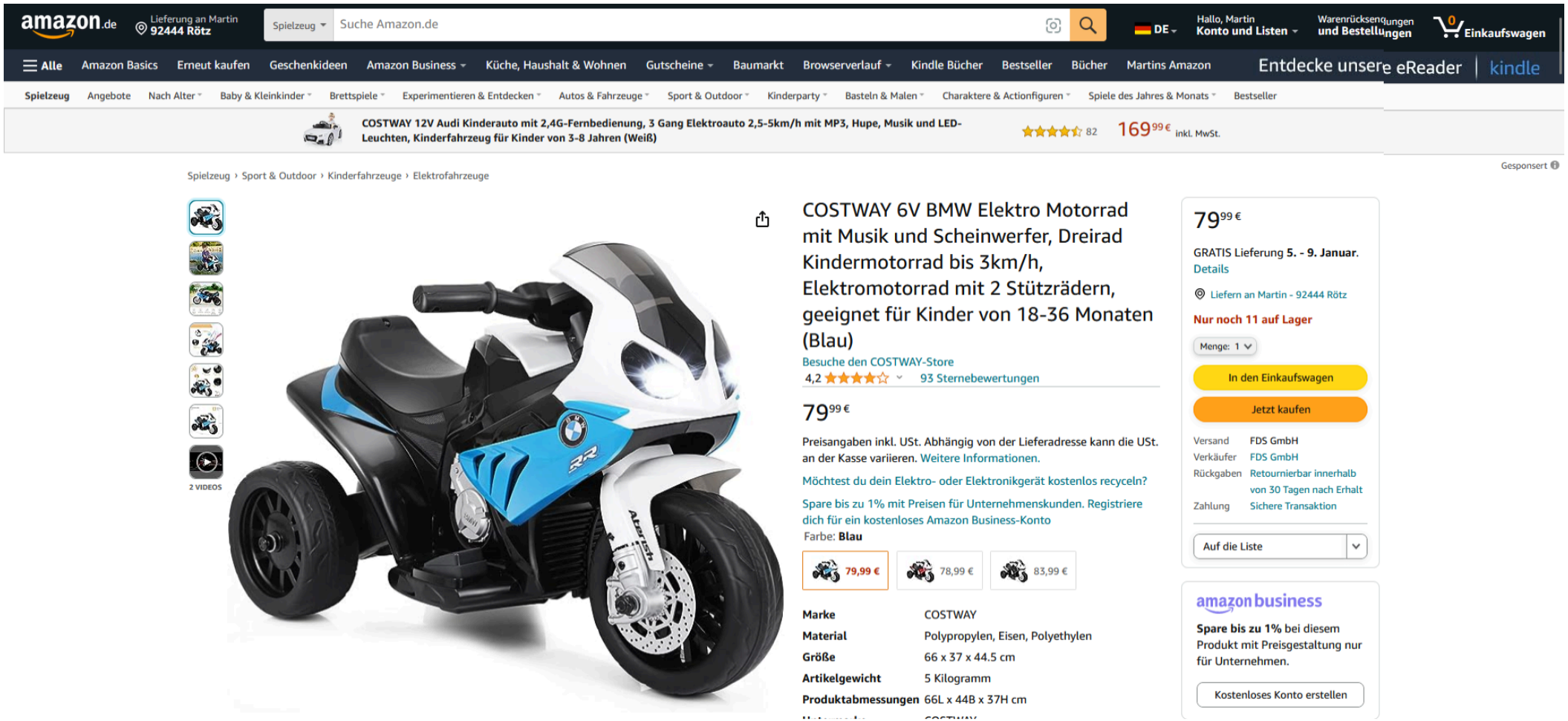

ANNUAL REPORT
29,268
2.06


Pass notiens
Personal Information
Medical History
Medications
Notes

# Motivation


Booking.com
EUR
Ihre Unterkunft anmelden
Registrieren
Anmelden
Aufenthalte
Flüge
Flug + Hotel
Mietwagen
Attraktionen
Flughafentaxis
Startseite › Deutschland › Berlin (Bundesland) › Berlin › Treptow-Köpenick › Müggelseeperle (Hotel), Berlin (Deutschland) Angebote
Wir bieten den gleichen Preis
Suche
Reiseziel/Name der Unterkunft:
Berlin
Check-in-Datum
Donnerstag, 21. März 2024
Check-out-Datum
Samstag, 23. März 2024
2 Nächte
2 Erwachsene · 0 Kinder · 1 Zi...
Ganze Unterkünfte & Wohnungen
Ich reise geschäftlich
Suche
Übersicht
Info & Preise
Ausstattung
Hausregeln
Gästebewertungen (493)
Level 1 bei „Travel Sustainable"
Reservieren
Müggelseeperle
Müggelheimer Damm 145, Treptow-Köpenick, 12559 Berlin, Deutschland –
Tolle Lage – Karte anzeigen
Wir bieten den gleichen Preis
Ansprechend
493 Bewertungen
6,9
6,9
Hotelpersonal
8,4

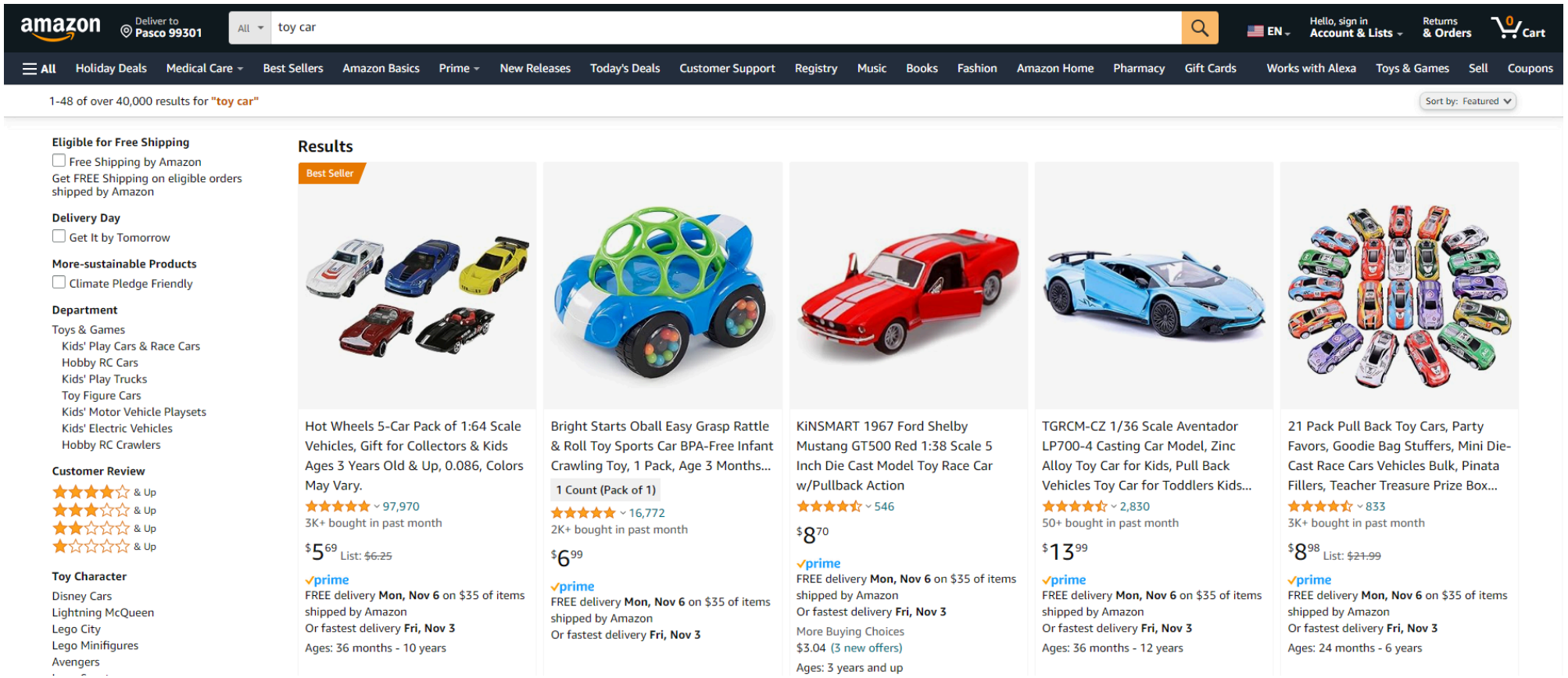
# Motivation


amazon.de
Lieferung an Martin
92444 Rötz
Spielzeug
Suche Amazon.de
DE
Hallo, Martin
Konto und Listen
Warenrücksendungen
und Bestellungen
Einkaufswagen
Alle
Amazon Basics
Erneut kaufen
Geschenkideen
Amazon Business
Küche, Haushalt & Wohnen
Gutscheine
Baumarkt
Browserverlauf
Kindle Bücher
Bestseller
Bücher
Martins Amazon
Entdecke unsere eReader
kindle
Spielzeug
Angebote
Nach Alter
Baby & Kleinkinder
Brettspiele
Experimentieren & Entdecken
Autos & Fahrzeuge
Sport & Outdoor
Kinderparty
Basteln & Malen
Charaktere & Actionfiguren
Spiele des Jahres & Monats
Bestseller
COSTWAY 12V Audi Kinderauto mit 2,4G-Fernbedienung, 3 Gang Elektroauto 2,5-5km/h mit MP3, Hupe, Musik und LED-Leuchten, Kinderfahrzeug für Kinder von 3-8 Jahren (Weiß)
82
169 99 € inkl. MwSt.
Gesponsert
Spielzeug › Sport & Outdoor › Kinderfahrzeuge › Elektrofahrzeuge
2 VIDEOS
COSTWAY 6V BMW Elektro Motorrad mit Musik und Scheinwerfer, Dreirad Kindermotorrad bis 3km/h, Elektromotorrad mit 2 Stützrädern, geeignet für Kinder von 18-36 Monaten (Blau)
Besuche den COSTWAY-Store
4,2
93 Sternebewertungen
79 99 €
Preisangaben inkl. USt. Abhängig von der Lieferadresse kann die USt. an der Kasse variieren. Weitere Informationen.
Möchtest du dein Elektro- oder Elektronikgerät kostenlos recyceln?
Spare bis zu 1% mit Preisen für Unternehmenskunden. Registriere dich für ein kostenloses Amazon Business-Konto
Farbe: Blau
79,99 €
78,99 €
83,99 €
Marke COSTWAY
Material Polypropylen, Eisen, Polyethylen
Größe 66 x 37 x 44.5 cm
Artikelgewicht 5 Kilogramm
Produktabmessungen 66L x 44B x 37H cm
79 99 €
GRATIS Lieferung 5. - 9. Januar.
Details
Liefern an Martin - 92444 Rötz
Nur noch 11 auf Lager
Menge: 1
In den Einkaufswagen
Jetzt kaufen
Versand FDS GmbH
Verkäufer FDS GmbH
Rückgaben Retournierbar innerhalb von 30 Tagen nach Erhalt
Zahlung Sichere Transaktion
Auf die Liste
amazon business
Spare bis zu 1% bei diesem Produkt mit Preisgestaltung nur für Unternehmen.
Kostenloses Konto erstellen

# Motivation

- Text and images contain important information for causal estimation.
- Here: Multimodal data as confounders
- Integrating of multimodal data into the Double / Debiased Machine Learning (DML)
- Applications in Marketing, Medicine / Health, Finance, ....
- Literature: Klaassen et al. (2024), Bach et al. (2023)

# Application



# Application: Estimation of Price Elasticity

- Understanding price elasticity is crucial for economic analysis and business decisions
  - It influences strategies, pricing, and market dynamics
- Traditional causal methods rely on structured data, limiting their applicability
- Utilizing unstructured data enhances decision-making
- Sources like images and text contain valuable insights
  - Contain rich hidden information about unobserved confounding variables

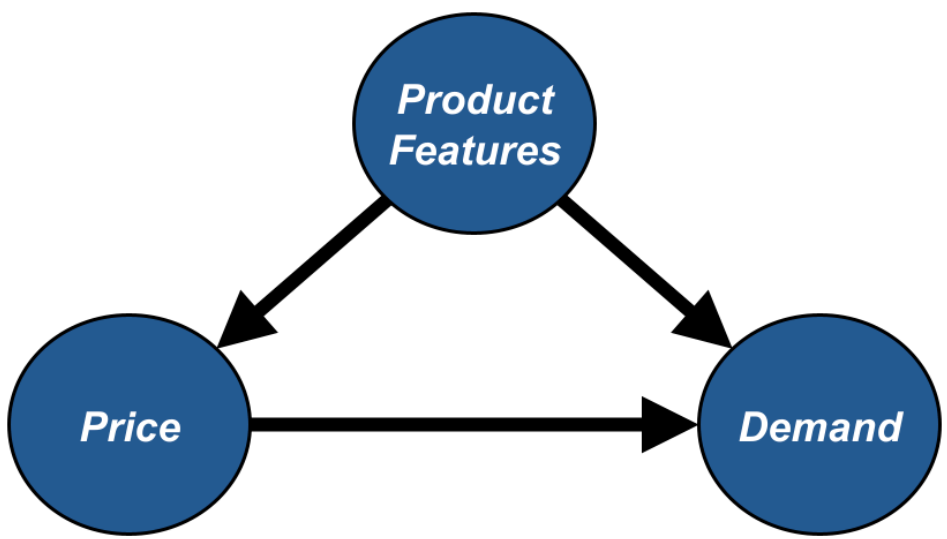
# Double Machine Learning

**Partially linear regression model (PLR)**

$$
\begin{aligned}
Y &= D\theta_0 + g_0(X) + \varepsilon, & \mathbb{E}[\varepsilon|P, X] &= 0, \\
D &= m_0(X) + V, & \mathbb{E}[V|X] &= 0,
\end{aligned}
$$

with

- Outcome $Y$, e.g. Demand $\ln(Q)$
- Treatment variable $D$, e.g. Price $\ln(P)$
- Additional product features $X = (X_1, \ldots)$
- Stochastic errors $\varepsilon$ and $V$




$\Rightarrow$ The causal parameter $\theta_0$ can be interpreted as price elasticity of demand!

# Double/Debiased Machine Learning (DML)

## Double/Debiased Machine Learning (DML)

- Use machine learning methods to fit

$$l_0(X) := \mathbb{E}[Y||X]$$
$$m_0(X) := \mathbb{E}[D|X]$$

- Construct the orthogonalized score (either cross-fitting or split the sample) for a fitted nuisance learner $\hat{\eta}_0 := (\hat{l}_0, \hat{m}_0)$

$$\psi(W, \theta, \hat{\eta}_0) := \Big(Y - \hat{l}_0(X) - \theta\big(D - \hat{m}_0(X)\big)\Big)\Big(D - \hat{m}_0(X)\Big)$$

- Compute the estimate as the solution of

$$0 = \frac{1}{n}\sum_{i=1}^{n} \psi(W, \hat{\theta}_0, \hat{\eta}_0)$$

# Double/Debiased Machine Learning (DML)

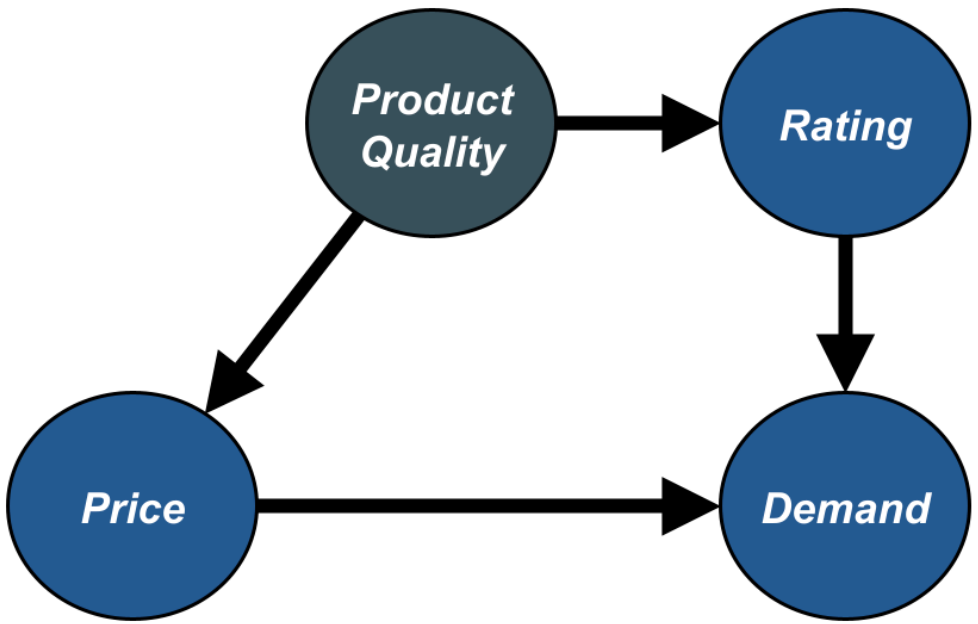
- Assumptions for the nuisance elements (see Chernozhukov et al. (2018))
  $\|\hat{m}_0 - m_0\|_{P,2} \times \left(\|\hat{m}_0 - m_0\|_{P,2} + \|\hat{\ell}_0 - \ell_0\|_{P,2}\right) \leq \delta_N N^{-1/2}$
- Under some regularity conditions the estimator $\hat{\theta}_0$ concentrates in $1/\sqrt{n}$-neighborhood of $\theta_0$ and

$$\sqrt{n}(\hat{\theta}_0 - \theta_0) \to \mathcal{N}(0, \sigma^2)$$

$\Rightarrow$ simple approach to control for product features, *if they can be used for ML models!*

# Application

- A presumably very important feature to consider might be product quality. How to determine product quality? Maybe use average rating, number of reviews, etc.

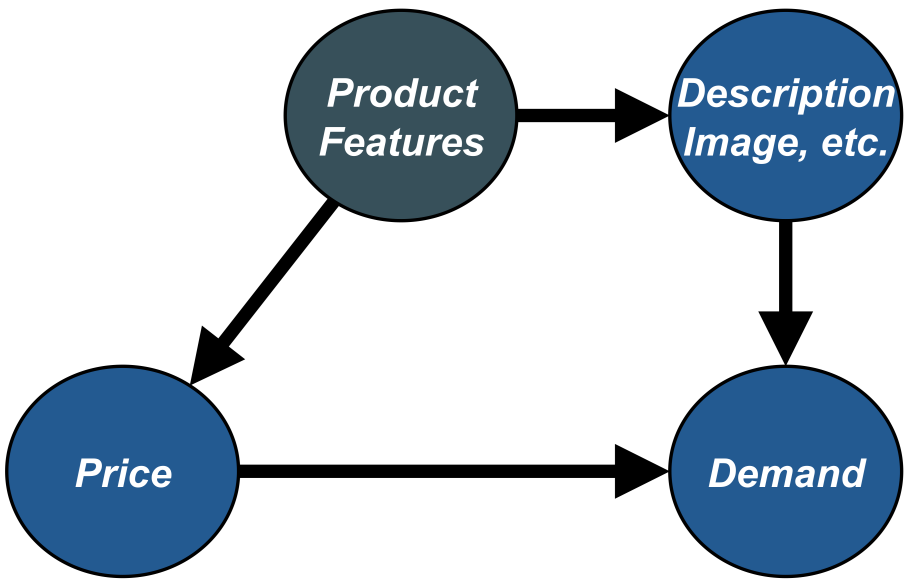



$\Rightarrow$ *Additionally rely on product description and images to describe the product feature!*

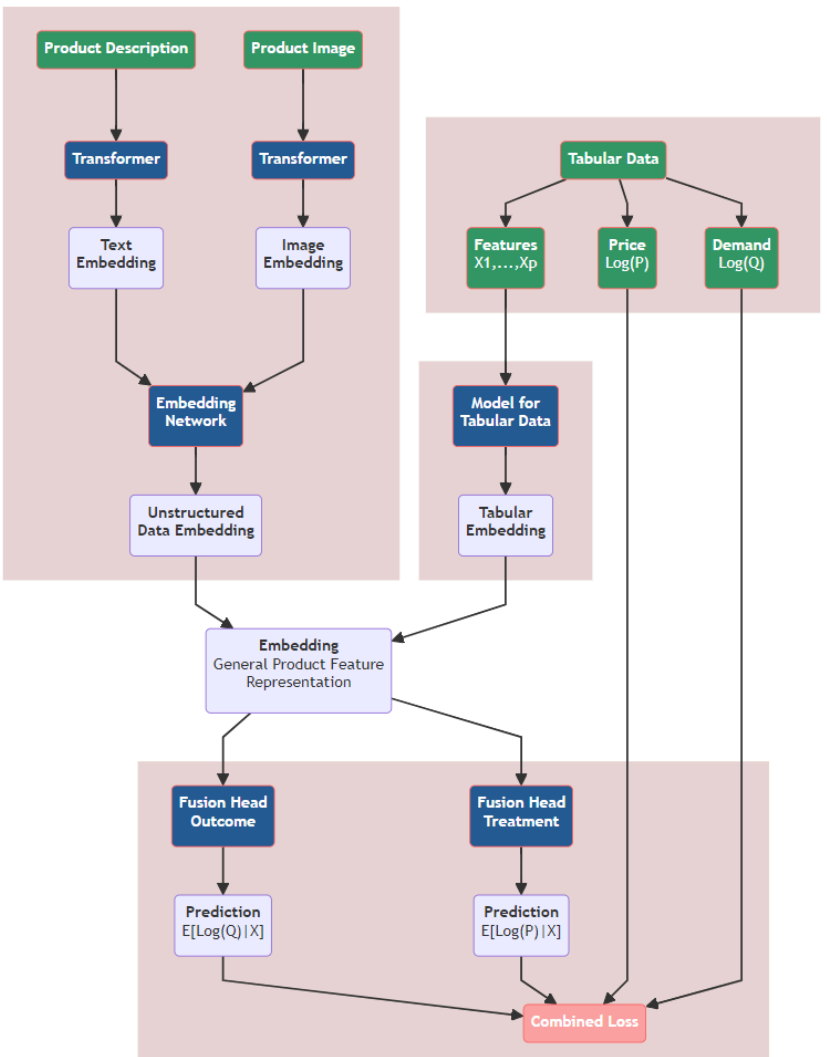
# DoubleML Deep

# DoubleML Deep - Motivation

- Use product description and image *additionally* to conventional tabular features

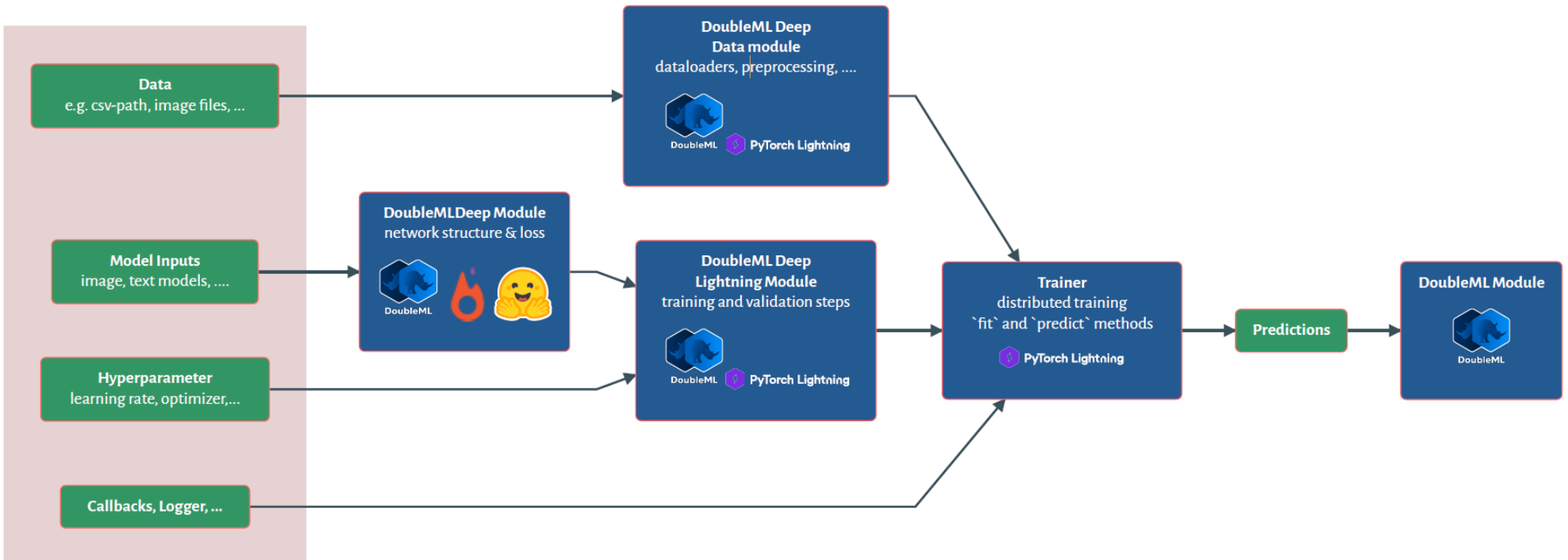


# DoubleML Deep - Module Structure




High-Level PLR Model Architecture. Both nuisance components are trained simultaneously with a combined loss.

# DoubleML Deep - Workflow




High-Level Workflow of *DoubleMLDeep*

⇒ More details can be found in Klaassen et al. (2024)

# DoubleML Deep - More Details

- Use pretrained models for unstructured data e.g. Bert and Beit
- Basic models for tabular data seem sufficient
- Monitor the nuisance losses
  - $\|\hat{l}_0 - \ln(Q)\|_{P,2} \leq \|\hat{l}_0 - l_0\|_{P,2} + \sigma_{\tilde{\epsilon}}$
  - $\|\hat{m}_0 - \ln(P)\|_{P,2} \leq \|\hat{m}_0 - m_0\|_{P,2} + \sigma_V$

# Application

# Amazon Toys Dataset

- Public Data from the Amazon sales platform from category toys and games for subcategories vehicles, cars and trucks.
- Using the Sales Rank as a proxy for quantities, as shown in Bajari et al. (2023).

| Variable | Description |
|---|---|
| Sales Rank | Sales rank as weighted mean for the last 30 days |
| Price | Price as weighted mean for the last 30 days |
| Text | Combination of Title, Category, Description, etc. |
| Image | First Image of the product |

# Amazon Toys Dataset

### Continous Variables

- Reviews: Rating
- Reviews: Review Count
- New Offer Count: Current
- Count of retrieved live offers: New, FBA
- Count of retrieved live offers: New, FBM

### Categorical Variables

- Lightning Deals: Upcoming Deal
- Buy Box: Is FBA

# Amazon Toys Dataset

## Data Examples

Kinsmart Set of 4 McLaren 720s Toy | ...



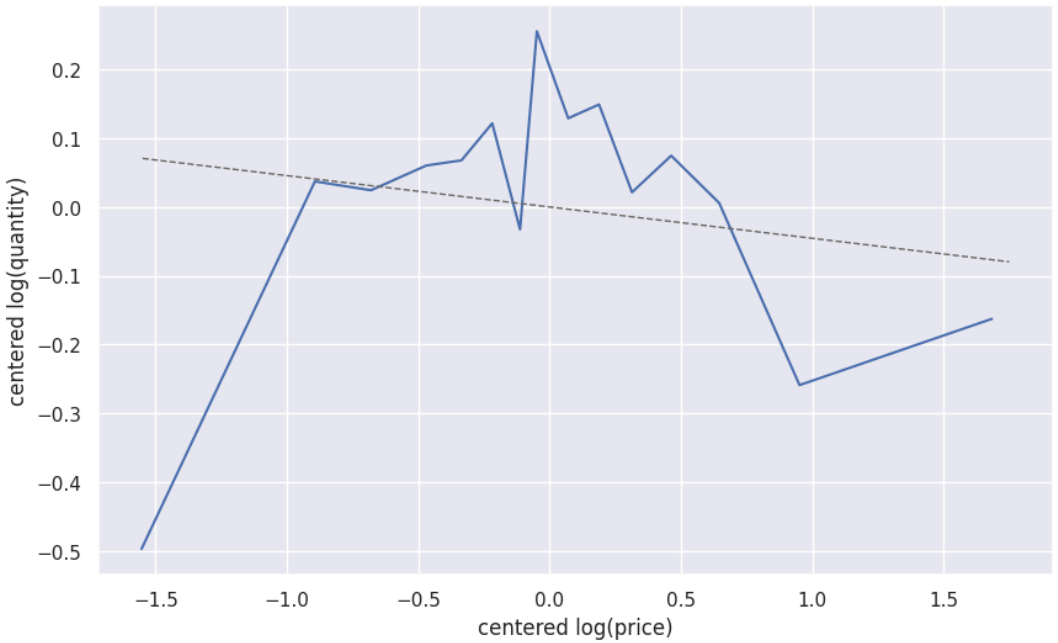
Pixar Cars Mack Uncle Lightning...



# Baseline

- Baseline estimate with tabular covariates, OLS model for regression
  - $\ln(Q) = \theta_0 \ln(P) + \beta^T X + \epsilon$
    - $R^2 = 0.330$

| 2.5 % | $\theta$ | 97.5 % |
|---|---|---|
| -0.072 | -0.046 | -0.019 |

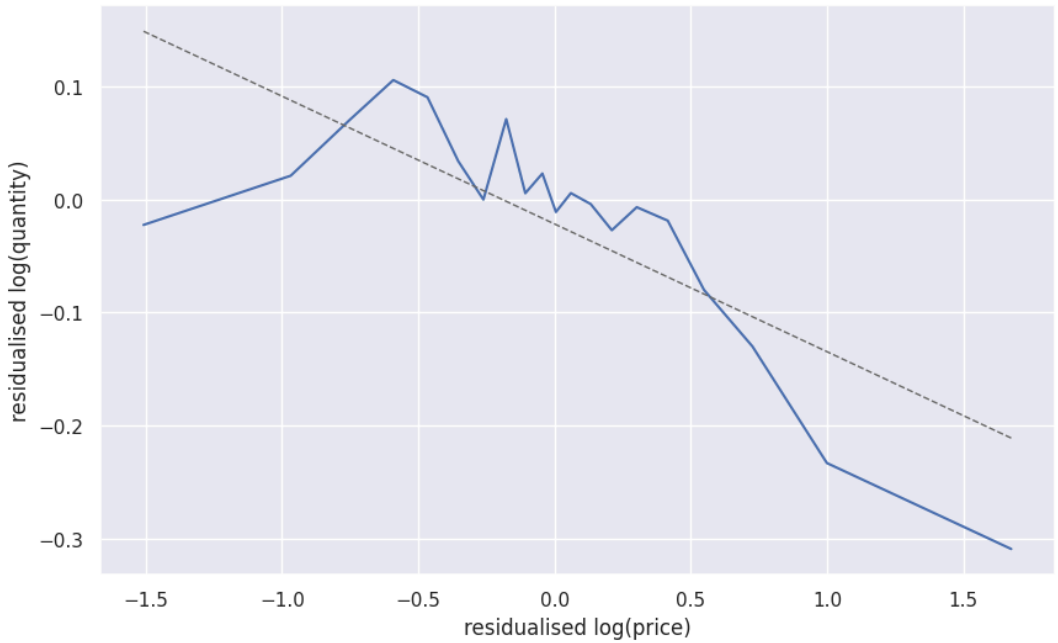



Centered logarithmic negative sales rank over centered logarithmic price as binned plot

# Baseline

- Estimate with tabular data as covariates $X$ in `DoubleMLPLR` model with `RandomForest` regressors
  - $l_0(X) := \mathbb{E}[\ln(Q)|X]$
    - $R^2_{l_0} = 0.5986$
  - $m_0(X) := \mathbb{E}[\ln(P)|X]$
    - $R^2_{m_0} = 0.1884$

| **2.5 %** | $\theta$ | **97.5 %** |
|---|---|---|
| -0.132 | -0.1098 | -0.080 |




Residualised logarithmic negative sales rank over residualised logarithmic price as binned plot

# Deep Learning Models

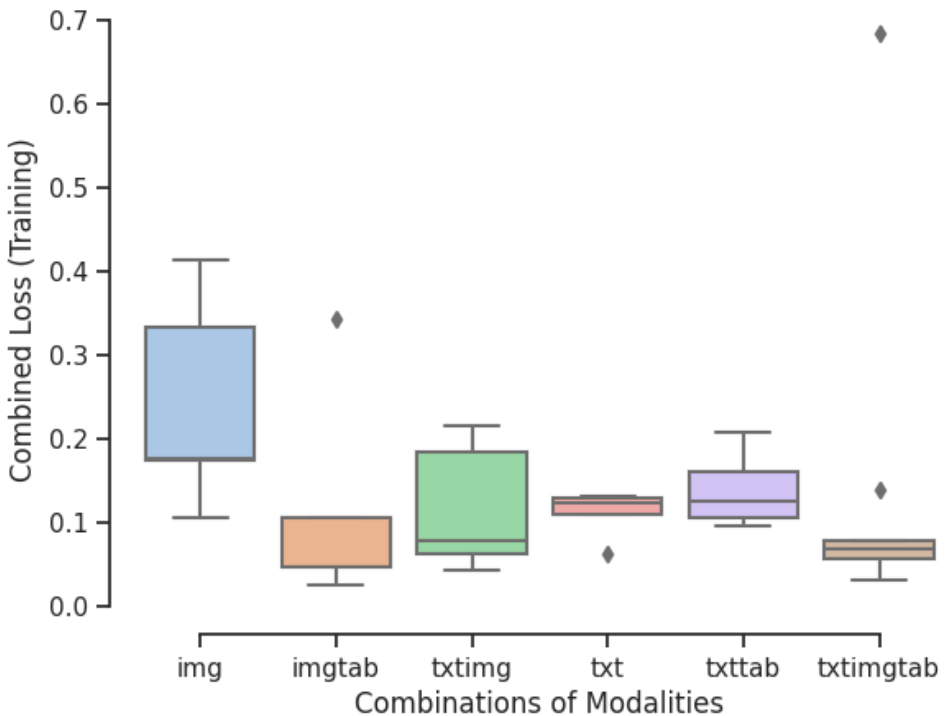
- In this study, a variety of input data combinations were used to train `DoubleMLPLRDeep` models [1]:

| Key | Used Confounders |
|---|---|
| `img` | $X = (X_{img})$ |
| `imgtab` | $X = (X_{img}, X_{tab})$ |
| `txtimg` | $X = (X_{txt}, X_{img})$ |
| `txt` | $X = (X_{txt})$ |
| `txttab` | $X = (X_{txt}, X_{tab})$ |
| `txtimgtab` | $X = (X_{txt}, X_{img}, X_{tab})$ |

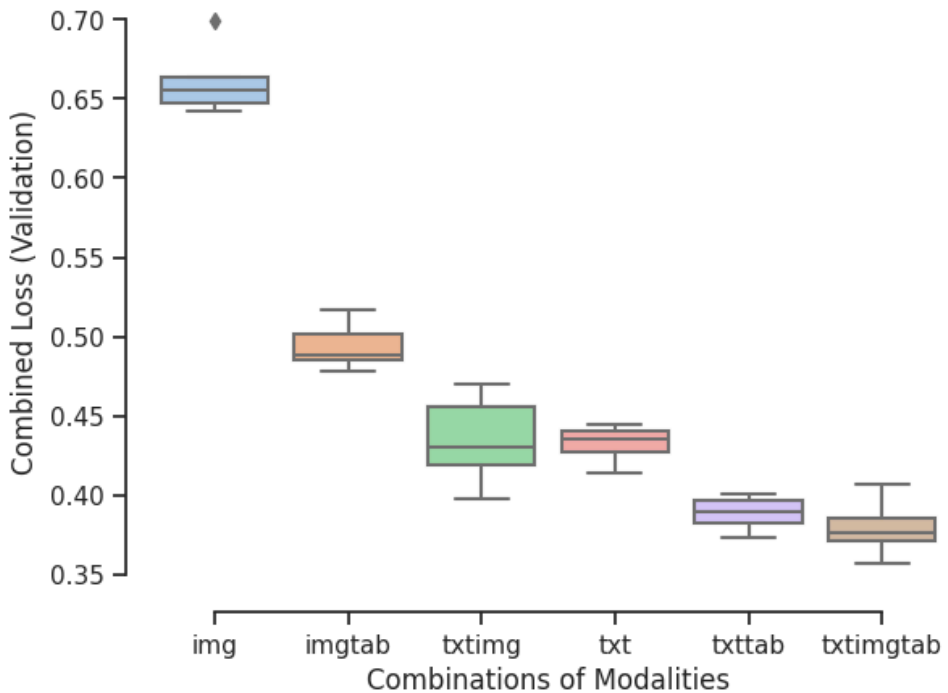
$\Rightarrow$ The aim is to emphasize the benefits of utilizing unstructured data.

# Deep Learning Models: RMSE-Scores

- Combined RMSE Score (Training)

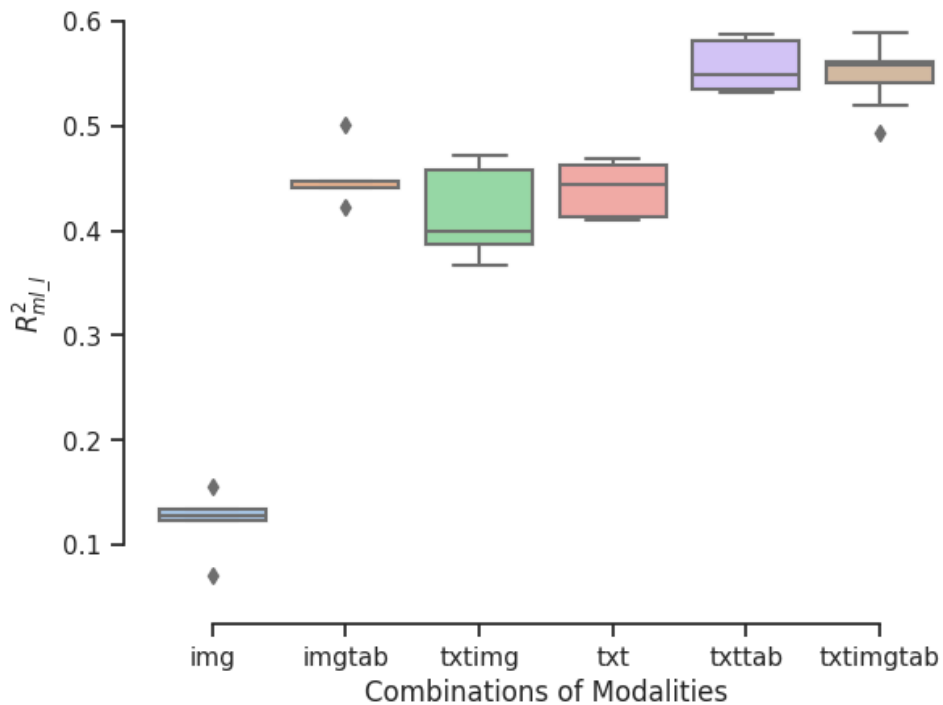



- Combined RMSE Score (Validation)

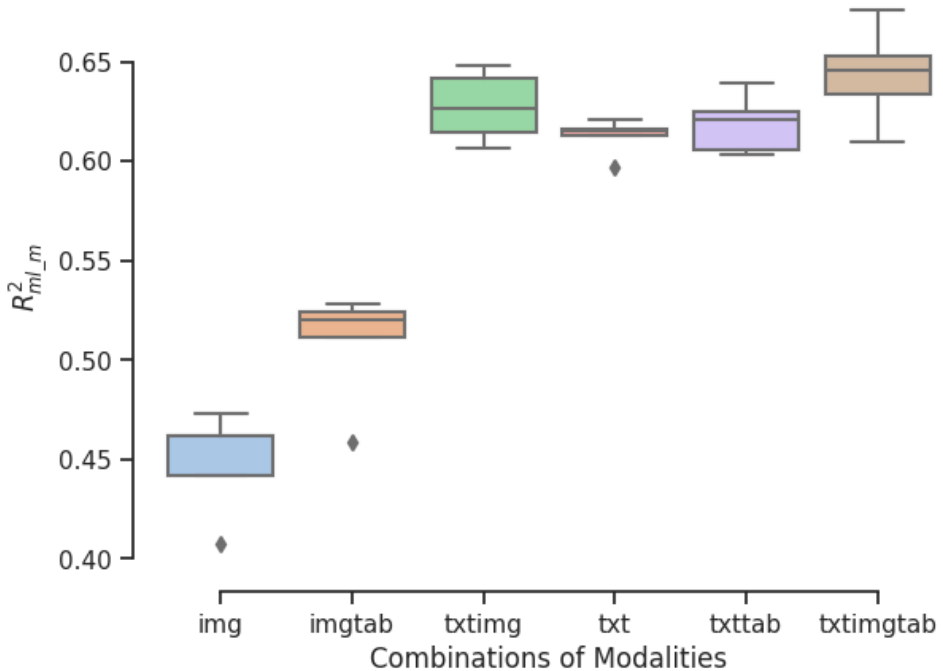


# Deep Learning Models: $R^2$-Scores

- $R^2$ of log(Quantity) on Validation Set

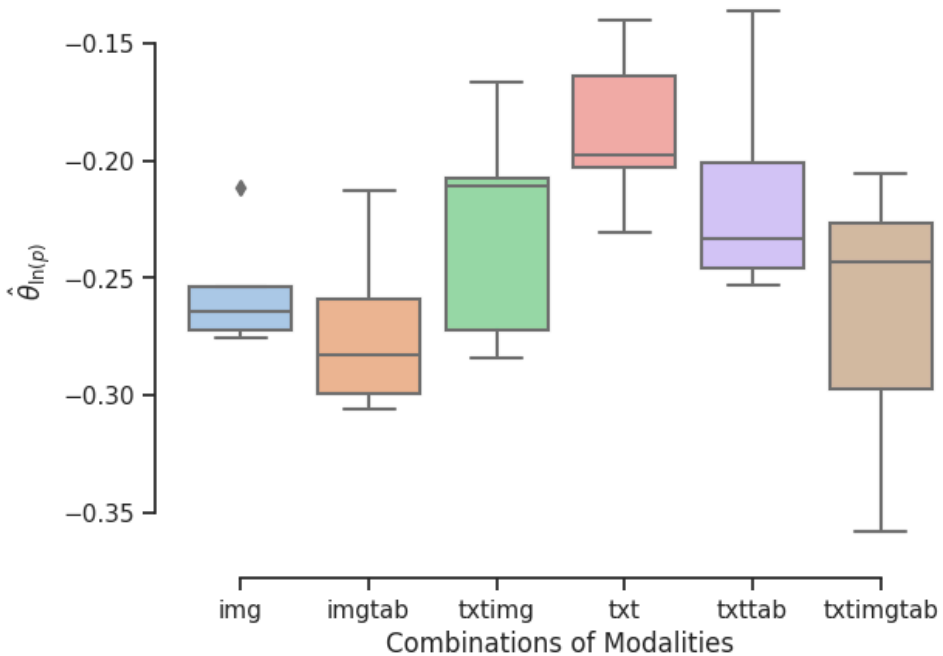


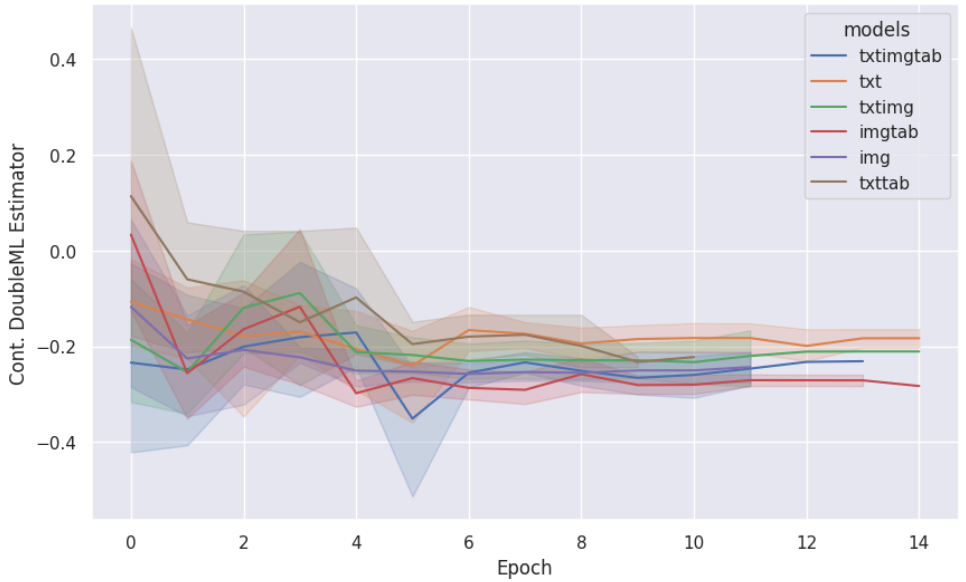

- $R^2$ of log(Price) on Validation Set



# (First) Results

- Using different combinations of tabular, image and text data we obtain the following estimates



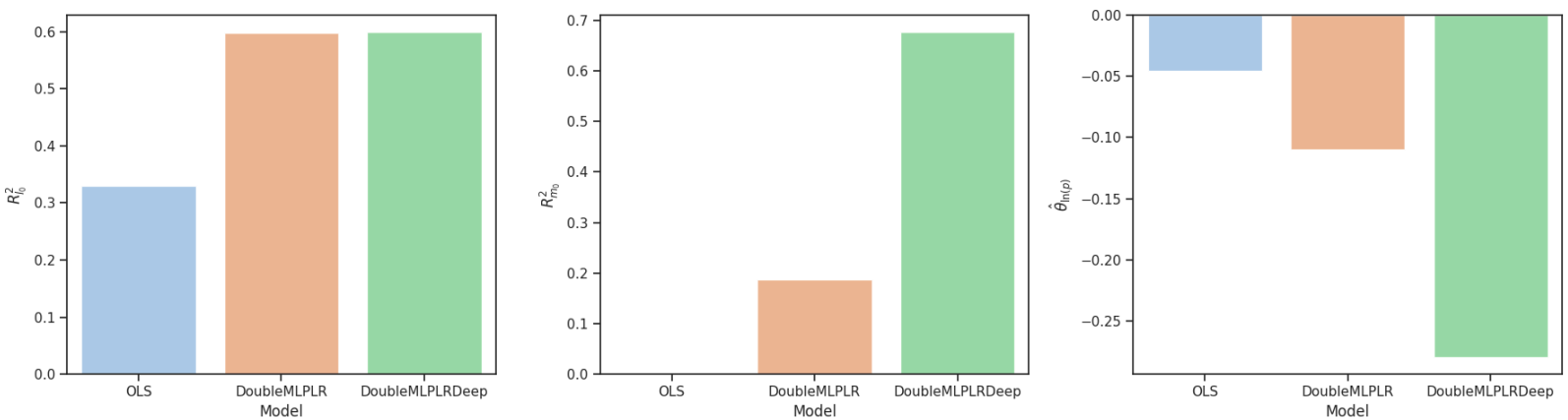




Continous Estimates from `DoubleMLPLR` on Validation Set during the Training Process

# Comparison to Baseline Estimates

| Model | Covariates | $R^2_{l_0}$ | $R^2_{m_0}$ | $\hat{\theta}_0$ |
|---|---|---|---|---|
| `OLS` | $X = X_{tab}$ | $0.3300$ | - | $-0.0455$ |
| `DoubleMLPLR` | $X = X_{tab}$ | $0.5986$ | $0.1884$ | $-0.1098$ |
| `DoubleMLPLRDeep` | $X = (X_{tab}, X_{txt}, X_{img})$ | $0.5990$ | $0.6765$ | $-0.2794$ |

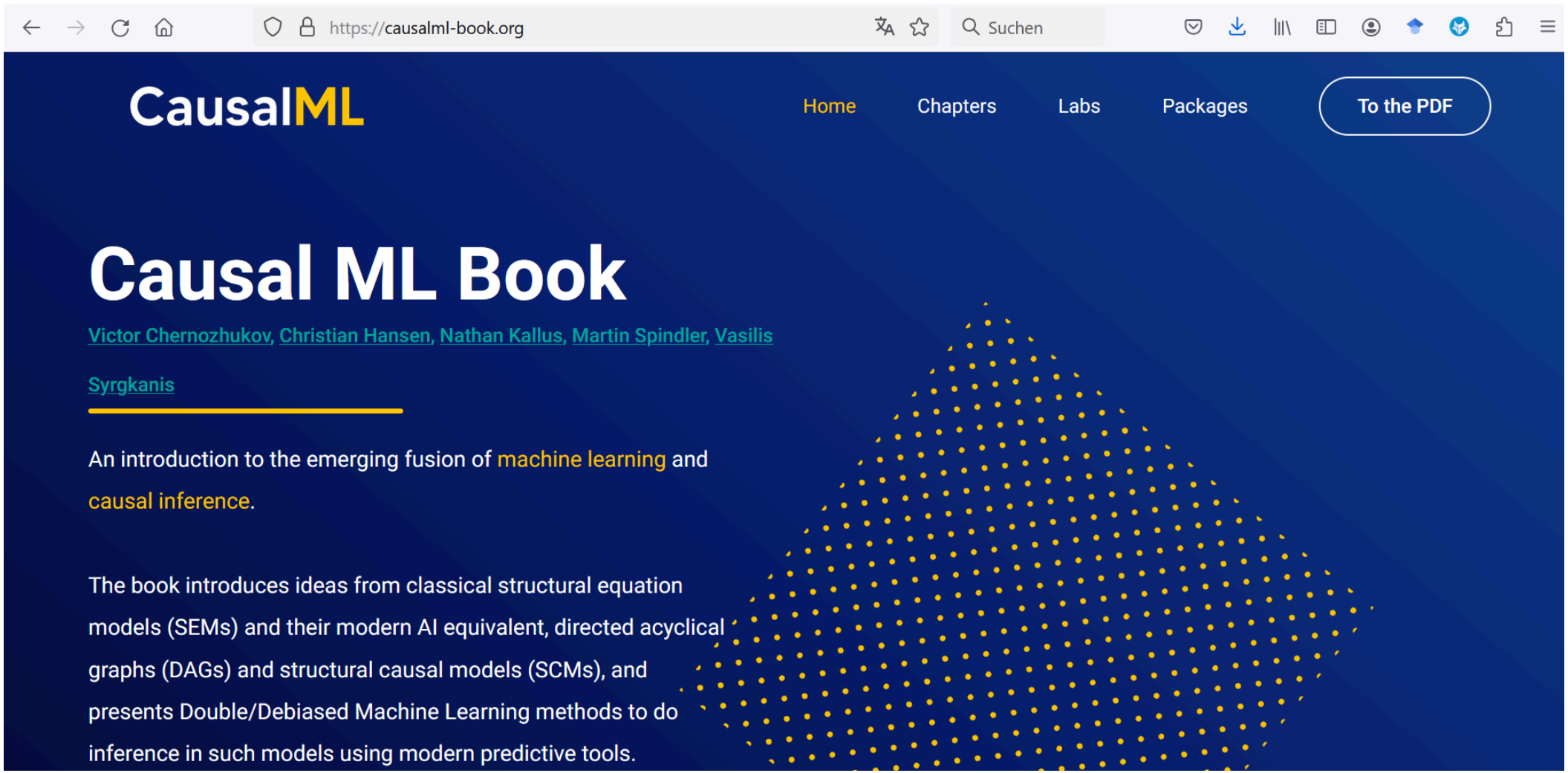
1



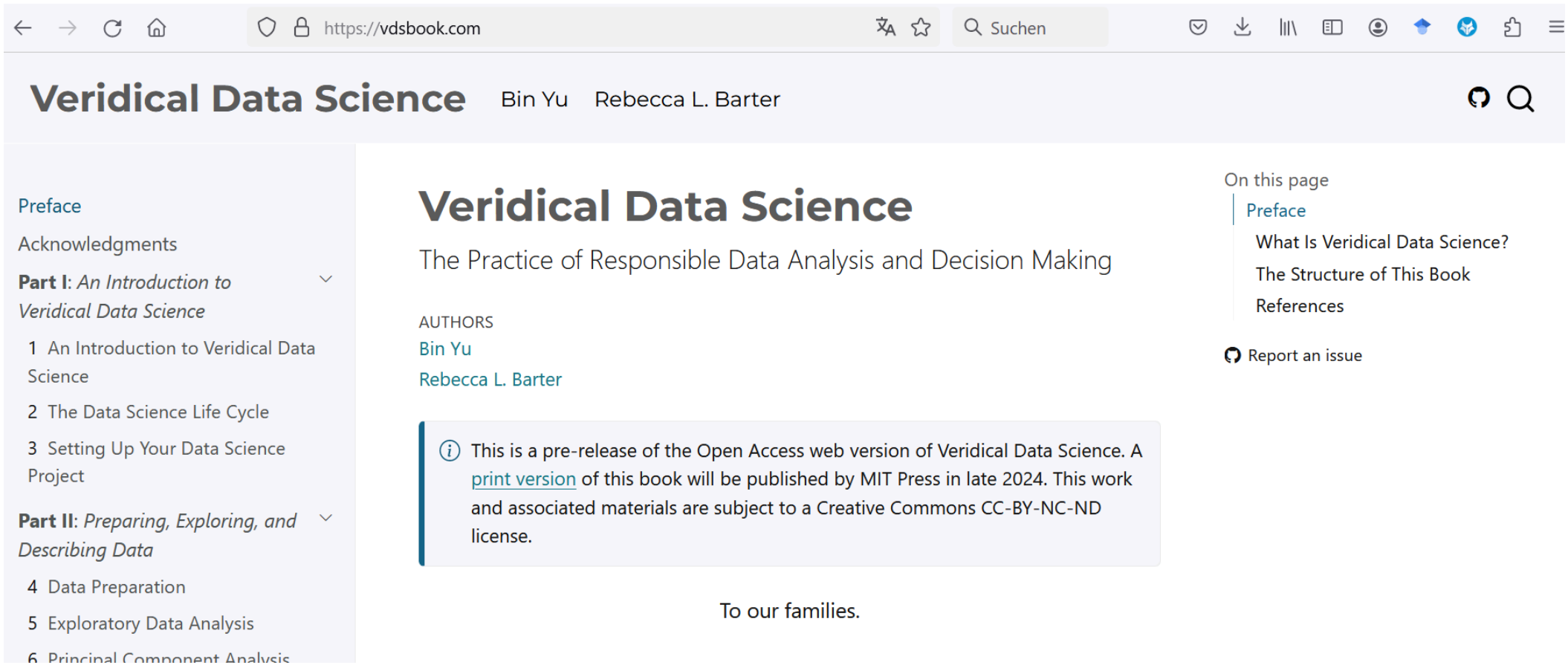
# Outline

- Semi-synthetical data for evaluation
- Further applications (Finance, Medicine)
- Extensions of the DoubleML package
- Using multiple time periods and controlling for unobserved confounder

# Double Machine Learning



# Veridical Data Science



# References

# References


Bach, Philipp, Victor Chernozhukov, Sven Klaassen, Martin Spindler, and Jan Teichert-Kluge. 2023. “DoubleML Deep - Estimating Price Elasticities.” In. University of Hamburg, Enonomic AI.

Bach, Philipp, Victor Chernozhukov, Malte S Kurz, and Martin Spindler. 2022. “DoubleML-an Object-Oriented Implementation of Double Machine Learning in Python.” *Journal of Machine Learning Research* 23: 53–51.

Bajari, Patrick, Zhihao Cen, Victor Chernozhukov, Manoj Manukonda, Suhas Vijaykumar, Jin Wang, Ramon Huerta, et al. 2023. “Hedonic Prices and Quality Adjusted Price Indices Powered by AI.” https://arxiv.org/abs/2305.00044.

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. “Double/Debiased Machine Learning for Treatment and Structural Parameters.” *The Econometrics Journal* 21 (1): C1–68. https://onlinelibrary.wiley.com/doi/abs/10.1111/ectj.12097.

Klaassen, Sven, Jan Teichert-Kluge, Philipp Bach, Victor Chernozhukov, Martin Spindler, and Suhas Vijaykumar. 2024. “DoubleMLDeep: Estimation of Causal Effects with Multimodal Data.” https://arxiv.org/abs/2402.01785.

# Affiliations




University of Hamburg

https://www.bwl.uni-hamburg.de/en/statistik/allgemein.html

**Thanks for your Attention!**